

IsoDecipher — Complete Scientific Framework & Roadmap

The Central Question

“Is APA switching a bifurcation event, and does this distinguish normal differentiation from oncogenic transformation?”

Part 1: What is IsoDecipher?

A targeted isoform quantification tool for **3' scRNA-seq data** that recovers:

- Alternative Polyadenylation (APA) — which polyA site does each cell use?
- 3' UTR length dynamics — short vs long UTR per cell
- Proximal Usage Index (PUI) — per cell, per gene

Critical Limitation (be honest):

IsoDecipher sees **3' end only**:

-  polyA site usage, 3' UTR length, APA
-  internal exon skipping, cassette exons, full-length isoforms
-  PUI is based on genomic coordinate UTR, not complete transcript

This is a feature too — GTF-anchored assignment is more accurate and less noisy than de novo APA tools (scAPA, Sierra).

Part 2: The Biological Observations (Your Data)

IGHM — Gradual shift (NOT bifurcation)

Naive B cell: G1 (membrane) dominant

Activated B cell: G0 + G1 coexist — secreted increases gradually

Plasma cell: G0 (secreted) dominant, G1 still present

Why gradual? Plasma cells still need some membrane IgM for BCR signaling. Secreted form increases but membrane form doesn't disappear. → APA switching is **reversible and gradual**

IGHG — Bifurcation-like (commitment event)

Memory B cell: G1 (membrane) only
Class switch →
Plasma cell: G0 (secreted) completely dominant
G1 essentially disappears

Why bifurcation? This is a true commitment — class-switched plasma cells are terminally differentiated. The switch is irreversible. → APA switching is **irreversible, bifurcation-like**

CD44 — Complex, cancer-relevant

CD44 has 20+ isoforms but 3' scRNA-seq only sees UTR length.

- Normal activation: gradual PUI changes with cell cycle
- Cancer/EMT: PUI shifts correlate with invasive state
- May show bifurcation in EMT context

CD274 (PD-L1) — Cancer-specific interest

- Normal: activation-regulated, reversible
- Cancer: persistent high expression → possible bifurcation lock-in
- Key immunotherapy target — APA dynamics could affect mRNA stability

Part 3: The Theoretical Framework

Waddington Landscape

Cell states as valleys in a potential landscape:

$U(x)$ = potential (depth of valley)
 $dx/dt = -\nabla U(x) + \text{noise}$



Cell State	Landscape	PUI	Stability
Naive/stem	Shallow valley	Variable	Medium
Transitioning	Ridge/saddle	Bimodal, unstable	Lowest
Terminal differentiated	Deep valley	Low variance	Highest
Cancer migrating (EMT)	Between attractors	Fluctuating	Very low
Proliferating	Outside attractor	High noise	Low

Bifurcation Theory Applied to APA

A bifurcation = qualitative change in system behavior from small parameter change.

In PUI terms:

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Before bifurcation: unimodal PUI distribution
At bifurcation:      bimodal PUI distribution (two stable states)
After bifurcation:   unimodal PUI distribution (new committed state)
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Detection signature:

- Bimodality Coefficient (BC) > 0.555 at bifurcation point
- Shannon entropy peaks at transition
- PUI variance peaks then drops sharply

Markov Chain & Differentiation Entropy (D_{En})

Cells modeled as nodes in a graph. Transition probabilities estimated from RNA velocity or expression similarity.

$$D_{En}(i) = -\sum_j T(i,j) * \log(T(i,j))$$

- High D_{En} = many possible destinations = progenitor/transitioning
- Low D_{En} = one clear destination = committed/terminal

Key insight — combining D_{En} with APA:

D _{En}	APA variance	Interpretation
Low	Low	TRUE terminal cell (deep attractor)

High	High	TRUE transitioning cell (saddle point)
Low	High	Fate committed, APA still resolving (lag!)
High	Low	APA committed first, fate still plastic

The last two quadrants reveal **temporal ordering** of molecular events:

- Which comes first — fate commitment or APA switching?
- This is a testable, original hypothesis.

Part 4: The Central Hypothesis

Gene-specific APA bifurcation behavior:

Not all genes behave the same way.

Classification:

Class	Example Genes	Biology	Reversible?
Bifurcation	IGHG, CD274 in cancer	True commitment event	No
Gradual	IGHM, CD44 in normal	Continuous regulation	Yes
Bimodal stable	Some surface markers	Two coexisting states	Partially
Noise	Most genes	No APA regulation	N/A

The Cancer Hypothesis:

"Oncogenic APA switching is irreversible (bifurcation-like), while normal differentiation APA switching can be gradual or reversible."

Prediction:

- Normal B cell differentiation: IGHM = gradual, IGHG = bifurcation
- Cancer EMT: CD44, CD274 = bifurcation-like lock-in
- APA bifurcation could distinguish epigenetic cancer state from normal activation

Why this matters clinically:

- Bifurcation point = therapeutic window (intervene before commitment → redirect cell fate)
- APA bifurcation signature = early epigenetic cancer biomarker (detect cancer state before morphological changes)

Part 5: Information Theory Layer

Adding mathematical rigor to stability measurements:

Method	Measures	Application
Shannon Entropy	Diversity of APA usage	Cell state heterogeneity
JSD	Distribution distance	APA divergence between cell types
Mutual Information	Statistical coupling	Which TFs drive APA switching?
Transfer Entropy	Causal direction in time	Does APA precede or follow fate decision?
Total Correlation	3-modal shared info	Transcriptome + secretome + APA co-regulation

Unique to your data (SEC-seq + IsoDecipher): Total Correlation across all three modalities simultaneously — no other dataset or tool can do this.

Part 6: Dimensionality Reduction — Which Method When

Method	Linear?	Global structure	Pseudotime?	Use for
PCA	✓ Yes	✓ Good	Barely	Initial exploration
Diffusion Map	✗ No	✓ Good	✓ Designed for this	Pseudotime calculation
UMAP	✗ No	⚠ Partial	⚠ Not reliable	Visualization

t-SNE	✗ No	✗ Poor	✗ No	Rare cell populations
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Diffusion Map math (intuition):

- Build KNN graph from cell similarity
- Model random walk on graph
- Pseudotime = diffusion distance from root cell
- Non-linear because distance follows data manifold, not straight line

For your data:

- No velocity data → use Diffusion Pseudotime (DPT)
- Have velocity data → use CellRank + scVelo latent time

Part 7: Data Strategy

Your own data:

- Ex vivo B cell culture → plasma cell differentiation
- Best for: IGHM, IGHG APA dynamics
- Limitation: ex vivo culture ≠ in vivo biology

Public cancer datasets (for oncology application):

- **GSE139555** — Tumor infiltrating B cells, multiple cancers
- **GSE181061** — Breast cancer single-cell
- **10x Genomics public** — processed h5ad, ready to use
- Key genes: CD274 (PD-L1), CD44, MYC, ITGA series

BAM files:

- Check if Cell Ranger BAM has CB + UB tags
- If yes → run velocityto → get spliced/unspliced → scVelo
- If only processed counts → use DPT instead

Part 8: Paper Story

Title options:

- 1. *"APA switching is a gene-specific bifurcation event: single-cell evidence from IsoDecipher"*
- 2. *"Reconstructing the Waddington landscape from APA dynamics in single-cell RNA-seq"*
- 3. *"IsoDecipher: information-theoretic dissection of APA dynamics distinguishes normal differentiation from oncogenic transformation"*

Figure roadmap:

- 1. **Figure 1** — IsoDecipher tool overview, IGHM example
- 2. **Figure 2** — APA trajectory along pseudotime (B cell differentiation)
- 3. **Figure 3** — Bifurcation detection: IGHM (gradual) vs IGHG (bifurcation)
- 4. **Figure 4** — Waddington landscape reconstruction from PUI
- 5. **Figure 5** — Combined stability score (DEn + APA variance) quadrant
- 6. **Figure 6** — Cancer application: CD274, CD44 APA in tumor microenvironment
- 7. **Figure 7** — Transfer entropy: temporal ordering of APA vs fate commitment

Part 9: Implementation Roadmap (週一開始)

Week 1 — Core analysis

Day	Task
Monday	PUI notebook — basic downstream analysis
Tuesday	APA trajectory — pseudotime + smoothing + JSD
Wednesday	Bifurcation detection — bimodality coefficient
Thursday	Waddington landscape visualization
Friday	Gene classification (bifurcation vs gradual vs noise)

Week 2 — Advanced analysis

Day	Task
Monday	CellRank integration + DEn computation
Tuesday	Combined stability quadrant (DEn + APA variance)
Wednesday	Information theory metrics (MI, Transfer Entropy)
Thursday	Public cancer dataset download + IsoDecipher application
Friday	Total Correlation (transcriptome + secretome + APA)

Week 3 — Publication preparation

Day	Task
Monday-Tuesday	Check BAM files for velocyto compatibility
Wednesday	Package IsoDecipher (pyproject.toml, pip installable)
Thursday	GitHub tutorial notebooks
Friday	bioRxiv preprint draft outline

Part 10: Why This Is Competitive

What makes you unique:

1. **SEC-seq inventor** — only you have transcriptome + secretome simultaneously
2. **IsoDecipher** — APA at single-cell resolution
3. **ODE modeling background** — attractor theory is natural to you
4. **Flow cytometry weekly** — can validate computational predictions experimentally
5. **B cell biology expertise** — know exactly whatIGHM/IGHG switching means

What no one else can do:

Total Correlation across transcriptome + secretome + APA: → Requires SEC-seq (yours) + IsoDecipher (yours) → This analysis is impossible without both tools

The competitive algorithm development angle:

- Bifurcation detection in APA = novel
- Gene-specific APA classification = novel
- Combined DEn + APA stability score = novel
- Transfer entropy for temporal ordering of APA vs fate = novel

None of these exist in current tools.